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CLINICAL TRIALS WORKSHOP

Oral Diuretic Intensification as a Worsening Heart Failure Event in the Primary Outcome of Clinical Trials

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Abstract

Although evidence-based therapies for patients with heart failure (HF) have decreased all-cause mortality, the residual risk of other clinically relevant outcomes — such as worsening HF — remains high. In this context, contemporary HF trials have increasingly employed composite primary outcomes that include worsening HF, commonly defined as episodes resulting in hospitalization for HF or urgent ambulatory visits with use of intravenous HF pharmacotherapies. In this Clinical Trials Workshop, we propose that the definition of worsening HF in clinical trials be expanded to include HF episodes treated with ambulatory oral diuretic intensification (ODI). Using previously reported post hoc analyses of pharmacotherapy trials in chronic HF, we highlight the prognostic significance of ODI and examine the implications of including ODI on anticipated event rates and estimated treatment effect. We propose a standardized definition of ODI and discuss challenges and regulatory considerations of incorporating ODI into HF trial end points.

Introduction

With the implementation of evidence-based therapies, mortality rates in heart failure (HF) randomized controlled trials have decreased over time, but the residual risk of HF events has remained high.[1] To fully capture this risk, along with the benefit of new treatments, the primary outcome in pivotal HF pharmacotherapy trials has evolved from all-cause mortality to a composite of outcomes that include worsening HF events.[1] Traditionally limited to episodes resulting in hospitalization for HF, the definition of worsening HF in clinical trials has been broadened to include urgent ambulatory visits requiring intravenous (IV) therapies without hospitalization.[2]

In contrast to this historical formulation, worsening HF in clinical practice is typically managed initially with oral diuretic intensification (ODI).[3] Barriers to including ODI as a HF event in the primary outcome of trials may include uncertainty about its clinical relevance, concerns about its reliable capture in trial data forms, and perceptions that it may add "noise" to treatment effect estimates. Furthermore, a standard, evidence-informed definition of ODI is lacking.

In this Clinical Trials Workshop, we propose expanding the definition of worsening HF to include ODI. We discuss a curated group of pivotal HF pharmacotherapy randomized trials

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that collected information regarding ODI. Using previously reported primary and post hoc analyses, we highlight that ODI is associated with all-cause mortality and that its inclusion in trials could substantially increase event rates, generate consistent treatment effect estimates, and improve statistical efficiency. We review the varied definitions of ODI in chronic HF trials and propose a standardized definition. Finally, we frame challenges and regulatory considerations when including ODI in the primary outcome of HF trials.

Definition and Prognostic Significance

DEFINITION OF ODI

Worsening HF is often subacute, with increases in left ventricular end-diastolic pressure weeks before hospitalization. The gradual worsening of symptoms associated with this often responds to intensification of oral diuretics, precluding the need for hospitalization.[4] To date, most HF trials have not included ODI in the definition of worsening HF, focusing generally on IV diuretic use or HF hospitalization. When ODI has been evaluated in prior trials, its definition has been heterogeneous with variable requirements for symptoms or signs of HF, type of diuretic, magnitude of dose increment, and duration of dose intensification (Table 1).[5-18] For example, trials such as PARADIGM-HF (Prospective Comparison of ARNI [angiotensin receptor — neprilysin inhibitor] With ACE [angiotensin-converting enzyme] Inhibitor to Determine Impact on Global Mortality and Morbidity in Heart Failure)[11] and DAPA-HF (Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure)[6] defined ODI as the sustained initiation or increase in dose of oral diuretics for >4 weeks for worsening symptoms, whereas trials such as TOPCAT-Americas (Treatment of Preserved Cardiac Function Heart Failure with an Aldosterone Antagonist — Americas),[7] FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients with Heart Failure),[9] and VICTOR (Vericiguat in Patients with Chronic Heart Failure and Reduced Ejection Fraction)[17] did not require the sustained intensification of diuretics or the presence of HF symptoms. Unlike some older HF pharmacotherapy trials in which oral loop diuretics were specified in the ODI definition, more contemporary trials such as FINEARTS-HF[9] and VICTOR[17] also included thiazide diuretics in their ODI definition (Fig. 1 and Table 1). Similarly, trials in patients at risk of HF have varied in their definition of ODI (Fig. 1).[10,18]

PROGNOSTIC SIGNIFICANCE OF ODI

Regardless of the ODI definition used or the left ventricular ejection fraction category of HF evaluated, post hoc analyses of several HF trials suggest that ODI is associated with a higher risk of death (Table 1). For example, in post hoc analyses of trials enrolling patients with HF with reduced ejection fraction (HFrEF), (e.g., PARADIGM-HF,[11] DAPA-HF,[6] VICTOR[17]), ODI was associated with an increased risk of death relative to no HF event. Likewise, in post hoc analyses of trials enrolling patients with HF and mildly reduced ejection fraction (HFmrEF) or preserved ejection fraction (HFpEF) — such as TOPCAT-Americas,[7] DELIVER (Dapagliflozin Evaluation to Improve the Lives of Patients with Preserved Ejection Fraction Heart Failure),[8] and FINEARTS-HF[9] — ODI was associated with an increased risk of death relative to no HF event. The magnitude of mortality risk portended by ODI was similar to that observed for urgent HF treated with IV diuretics, as demonstrated in the DAPA-HF[6] and FINEARTS-HF trials.[9] Furthermore, in patients treated with ODI, larger diuretic dose increments were associated with a greater risk of death.[8,9]

Event Rates and Statistical Efficiency

ANTICIPATED EVENT RATES WITH ODI IN THE PRIMARY OUTCOME

An expanded definition of worsening HF that includes ambulatory ODI would be anticipated to increase event rates relative to traditional definitions restricted to HF hospitalization or IV diuretic use. We provide examples of event rates for ODI in a select group of HF pharmacotherapy trials across left ventricular ejection fraction categories in Table 1.[5-11,13,17,18]

Ambulatory presentations of worsening HF are common, and among these, treatment with oral diuretics is much more common than IV diuretics. This observed epidemiology of HF presentation could facilitate an increase in event rates with the inclusion of ODI in the primary outcome of HF trials. For example, in patients with HFrEF in the DAPA-HF trial, the first episode of worsening HF was a HF hospitalization in 10.3% of patients, an ambulatory event in 8.6% of patients, and an urgent visit with IV therapy in 0.4% of patients, highlighting the important potential contribution of ambulatory events to a trial's event rates.[6] In VICTOR, expanding the definition of worsening HF from HF hospitalization to a composite that included urgent HF

Table 1. Select Heart Failure Trials that Analyzed Oral Diuretic Intensification as an Outcome.*					
Trial, Sample Size, Comparator, and Follow-Up	**Definition of Oral Diuretic Intensification**	**Oral Diuretic Intensification Event Rates in the Placebo Group**	**Prognostic Implications of Diuretic Intensification**	**Treatment Effect on the Primary Composite Outcome**	**Treatment Effect on Relevant Exploratory Outcomes**
HFrEF RCTs					
PARADIGM-HF 8399 patients Sacubitril/valsartan vs. enalapril 27 months (median)	Initiation or increase in dose of diuretics sustained for >4 weeks for worsening symptoms	Data not provided	Increased risk of all-cause death (adjusted hazard ratio,† 3.2; 95% CI, 2.2 to 5.0) relative to no HF event	Primary composite outcome of CV death or hospitalization for HF: hazard ratio, 0.80 (95% CI, 0.73 to 0.87)	Expanded composite outcome of CV death, hospitalization for HF, emergency department visit for HF, or outpatient intensification of any oral or IV HF therapies (including diuretics): hazard ratio, 0.79 (95% CI, 0.73 to 0.86)
DAPA-HF 4744 patients Dapagliflozin vs. placebo 18.2 months (median)	Initiation or increase in dose of diuretics sustained for >4 weeks for worsening outpatient symptoms	11.9 events per 100 patients during follow-up§	Increased risk of all-cause death (adjusted hazard ratio,‡ 2.82; 95% CI, 2.11 to 3.77) relative to no HF event	Primary composite outcome of CV death or worsening HF, defined as HF hospitalization or urgent HF visit resulting in IV therapy: hazard ratio,¶ 0.74 (95% CI, 0.65 to 0.85); ARR, 4.0%	Expanded composite outcome of CV death or worsening HF or oral intensification of any HF therapy: hazard ratio,¶ 0.73 (95% CI, 0.65 to 0.82); ARR, 6.1%
EMPEROR-Reduced 3730 patients Empagliflozin vs. placebo 16 months (median)	Initiation or increase in oral daily dose of diuretics for worsening symptoms	19.4 events per 100 patient-years	Data not provided	Primary composite outcome of CV death or HF hospitalization: hazard ratio,‖ 0.75 (95% CI, 0.65 to 0.86; ARR, 5.2%	Expanded composite outcome of all-cause death, HF hospitalization, or urgent HF care requiring IV therapy: hazard ratio,‖ 0.76 (95% CI, 0.67 to 0.87); ARR, 5.5% Expanded composite outcome of all-cause death, HF hospitalization, emergent/urgent HF care requiring IV treatment, or diuretic intensification: hazard ratio,‖ 0.70 (95% CI, 0.63 to 0.78); ARR, 10.8%
VICTOR 6105 patients Vericiguat vs. placebo 18.5 months (median)	(1) Initiation of loop or thiazide diuretic; or (2) Intensification of oral loop or thiazide diuretic dose (relative to baseline)	20.5% of participants during follow-up§ (7.1% oral diuretic initiation, 13.4% oral diuretic dose intensification)	Increased risk of all-cause death (RR, 1.69; 95% CI, 1.47 to 1.94) relative to no intensification	Primary composite outcome of cardiovascular death or HF hospitalization: hazard ratio, 0.93 (95% CI, 0.83 to 1.04); ARR, 0.9%	Expanded composite outcome of cardiovascular death, HF hospitalization, urgent HF visit requiring IV diuretics, or oral diuretic initiation/intensification: hazard ratio, 0.89 (95% CI, 0.81 to 0.98); ARR, 2.3%
HFmrEF/HFpEF RCTs					
TOPCAT-Americas sub-analysis 1767 patients Spironolactone vs. placebo 2.9 years (median)	Initiation or increase in dose of oral loop diuretic (furosemide equivalent)	20.6 events per 100 patient-years	Increased risk of all-cause death (adjusted hazard ratio,** 1.75 (95% CI, 1.41 to 2.16) relative to no oral diuretic intensification	Primary composite outcome of CV death, aborted cardiac arrest, or HF hospitalization: hazard ratio, 0.82 (95% CI, 0.69 to 0.98); ARR, 2.2%	Composite outcome of CV death or HF hospitalization: hazard ratio, 0.84 (95% CI 0.70 to 0.99); ARR, 2.0% Expanded composite outcome of CV death, HF hospitalization, or oral diuretic intensification: hazard ratio, 0.74 (95% CI, 0.65 to 0.84); ARR, 8.4%
DELIVER 6263 patients Dapagliflozin vs. placebo 2.3 years (median)	Initiation or sustained increase in dose of oral loop diuretics for ≥30 days	18.8 events per 100 patients during follow-up§	All-cause death rate of 10 (95% CI, 8 to 12) per 100 patient-years in patients with oral diuretic intensification vs. 4 (95% CI, 3 to 4) per 100 patient-years in patients with no worsening HF	Primary composite outcome of CV death or worsening HF, defined as HF hospitalization or urgent HF visit with IV treatment: hazard ratio, 0.82 (95% CI, 0.73 to 0.92); ARR, 3.1%	Expanded composite outcome of CV death, worsening HF event, or oral diuretic intensification: hazard ratio, 0.76 (95% CI, 0.69 to 0.84); ARR, 6.2%

(*Continued*)

Table 1. (*Continued*)					
Trial, Sample Size, Comparator, and Follow-Up	**Definition of Oral Diuretic Intensification**	**Oral Diuretic Intensification Event Rates in the Placebo Group**	**Prognostic Implications of Diuretic Intensification**	**Treatment Effect on the Primary Composite Outcome**	**Treatment Effect on Relevant Exploratory Outcomes**
SUMMIT†† 731 patients Tirzepatide vs. placebo 104 weeks (median)	Worsening symptoms with (1) Initiation or increase in daily dose of oral diuretic; (2) An extra dose of oral diuretic; or (3) Resumption of oral diuretic in a patient in whom diuretic therapy was previously discontinued or reduced	3.2 events per 100 person-years	Data not provided	Primary composite outcome of CV death or a worsening HF event, defined as HF hospitalization, urgent HF visits treated with IV drugs, or oral diuretic intensification: hazard ratio,‡‡ 0.62 (95% CI, 0.41 to 0.95); ARR, 3.3%	Restricted composite outcome of CV death or worsening HF defined as HF hospitalization, urgent HF visits treated with IV drugs: adjusted hazard ratio,‡‡ 0.57 (95% CI, 0.34 to 0.95); ARR, 2.6%
FINEARTS-HF 6001 patients Finerenone vs. placebo 32 months (median)	(1) Initiation or increase in the dose of oral loop diuretics; or (2) Initiation of thiazide diuretics	27.8 events per 100 patients during follow-up§	All-cause death rate of 11.6 (95% CI, 10.2 to 13.1) per 100 patient-years in patients with oral diuretic intensification vs. 4.5 (95% CI, 4.2 to 4.9) per 100 patient-years in patients with no worsening HF event	Primary composite outcome of CV death or total worsening HF events, defined as HF hospitalizations or urgent HF visits: hazard ratio, 0.84 (95% CI, 0.74 to 0.95)	Composite outcome of CV death or worsening HF, defined as HF hospitalization or urgent HF visit: hazard ratio, 0.84 (95% CI, 0.76 to 0.94). Expanded composite outcome of CV death, worsening HF, or outpatient oral diuretic intensification: hazard ratio, 0.85 (95% CI, 0.78 to 0.92).

* ARR comparisons are presented for those trials that published or provided annualized event rates for exploratory outcomes. ACEi denotes angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARR, absolute risk reduction; ATTR, transthyretin amyloidosis; BMI, body-mass index; CI, confidence interval; CKD, chronic kidney disease; CRT, cardiac resynchronization therapy; CV, cardiovascular; DAPA-HF, Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure; DELIVER, Dapagliflozin Evaluation to Improve the Lives of Patients with Preserved Ejection Fraction Heart Failure; eGFR, estimated glomerular filtration rate; EMPEROR-Reduced, Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction; FINEARTS-HF, Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients with Heart Failure; HELIOS-B, Study to Evaluate Vutrisiran in Patients with Transthyretin Amyloidosis with Cardiomyopathy; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HR, heart rate; ICD, implantable cardioverter defibrillator; IV, intravenous; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; PARADISE-MI, Prospective ARNI versus ACE Inhibitor Trial to Determine Superiority in Reducing Heart Failure Events after Myocardial Infarction; SUMMIT, Study of Tirzepatide (LY3298176) in Participants with Heart Failure with Preserved Ejection Fraction and Obesity; RCT, randomized controlled trial; SBP, systolic blood presssure; TOPCAT-Americas, Treatment of Preserved Cardiac Function Heart Failure with an Aldosterone Antagonist — Americas; and VICTOR, Vericiguat in Patients with Chronic Heart Failure and Reduced Ejection Fraction.

† Adjusted for randomized treatment, region, and baseline covariates (age, sex, race, SBP, HR, BMI, serum creatinine, LVEF, NT-proBNP, NYHA functional class, ischemic etiology, hypertension, diabetes mellitus, atrial fibrillation, previous HF, myocardial infarction, stroke, previous ICD, and CRT).

‡ Stratified according to diabetes status, and adjusted for randomized treatment, age, sex, region, race, NYHA functional class, LVEF, BMI, HR, SBP, serum creatinine, log NT-proBNP, history of previous HF hospitalization, atrial fibrillation, stroke, myocardial infarction, hypertension, ischemic etiology of HF, and use of ICD and/or CRT.

§ The event rate per 100 patient-years was not provided.

¶ Stratified according to diabetes status and adjusted for a history of HF hospitalization and treatment-group assignment.

|| Adjusted for age, sex, geographical region, diabetes status at baseline, LVEF, and eGFR at baseline.

** Adjusted for age, sex, race, BMI, NYHA functional class, SBP, baseline potassium level, diabetes, eGFR, ACEi/ARB treatment, and loop diuretic use at baseline.

†† SUMMIT is the only trial in this table that included oral diuretic intensification in the primary composite outcome.

‡‡ Adjusted for diagnosed type 2 diabetes (yes/no), baseline probability of HFpEF (a HFpEF-ABA score of ≥0.8 or <0.8), and baseline NT-proBNP (<200, ≥200 pg/ml).

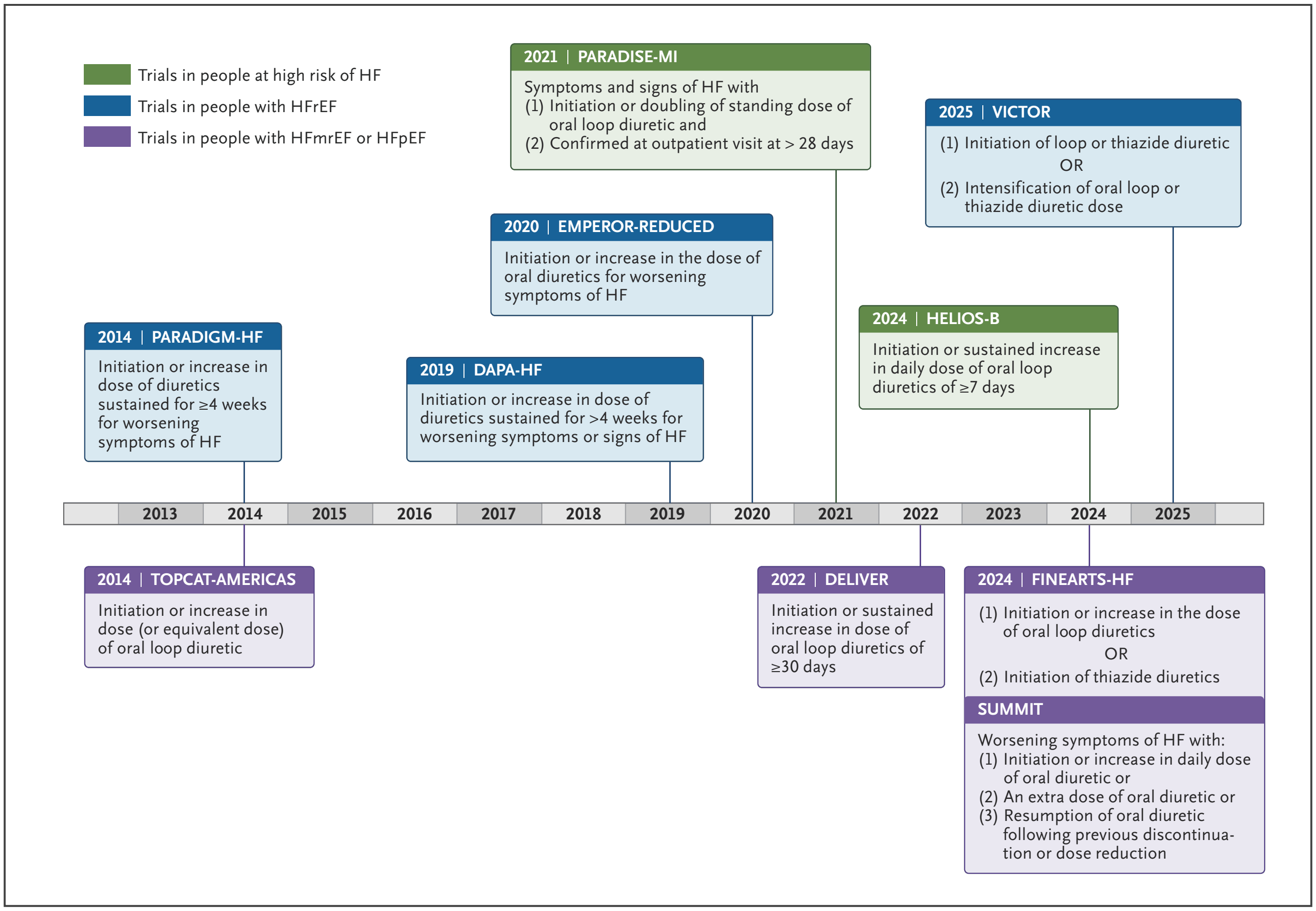

Figure 1. Definition of Oral Diuretic Intensification Used in Contemporary Trials.

ACE denotes angiotensin-converting enzyme; ARNI, angiotensin receptor — neprilysin inhibitor; DAPA-HF, Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure; DELIVER, Dapagliflozin Evaluation to Improve the Lives of Patients with Preserved Ejection Fraction Heart Failure; EMPEROR-Reduced, Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction; FINEARTS-HF, Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients with Heart Failure; HELIOS-B, Study to Evaluate Vutrisiran in Patients with Transthyretin Amyloidosis with Cardiomyopathy; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; PARADIGM-HF, Prospective Comparison of ARNI With ACE Inhibitor to Determine Impact on Global Mortality and Morbidity in Heart Failure; PARADISE-MI, Prospective ARNI versus ACE Inhibitor Trial to Determine Superiority in Reducing Heart Failure Events after Myocardial Infarction; SUMMIT, Study of Tirzepatide (LY3298176) in Participants with Heart Failure with Preserved Ejection Fraction and Obesity; TOPCAT, Treatment of Preserved Cardiac Function Heart Failure with an Aldosterone Antagonist; and VICTOR, Vericiguat in Patients with Chronic Heart Failure and Reduced Ejection Fraction.

visits with ODI (including initiation) or IV diuretics approximately doubled the HF events in the placebo group from 7.6 to 17.2 events per 100 patient-years.[17] Notably, in the VICTOR trial, at least 10 times as many patients with ambulatory HF events received treatment with ODI as IV diuretics. Among patients in the placebo group, the first HF event was most commonly ODI (14.9%), followed by HF hospitalization (8.2%), and urgent HF visit with IV diuretics (1.4%). Among those receiving oral loop diuretic intensification in the placebo group, 71.8% had dose increments (a majority had ≥50% dose increments), and 28.2% had dose initiation.[17]

In trials in patients with HFmrEF or HFpEF, the inclusion of ODI in a primary composite end point could also meaningfully increase event rates. For example, in TOPCAT-Americas, adding ODI to HF hospitalization would have increased the composite cardiovascular death or HF event rate in the placebo group by nearly threefold (from 10.4 to 29.9 events per 100 person-years).[7] Similarly, the inclusion of ODI in the definition of worsening HF would have increased the number of events in the placebo groups in DELIVER (from 610 to 962 events)[8] and in FINEARTS-HF (from 719 to 1189 events)[9] (Table 1).

IMPLICATIONS OF ODI INCLUSION ON STATISTICAL EFFICIENCY

The anticipated increase in event rates with the inclusion of ODI in the primary outcome could translate to greater statistical efficiency of HF trials, with smaller sample sizes or shorter durations of follow-up to accrue the number of events required to achieve statistical power. For example, in TOPCAT-Americas, adding ODI to the definition of a HF event would have reduced the mean time to event in the trial from 904 to 837 days.[7] In PARADIGM-HF, the time taken to accrue 1000 events would have decreased from 11 to 9 months had the primary composite outcome been expanded to include ODI.[11]

Estimated Treatment Effect

One concern regarding the inclusion of ODI in the primary outcome of trials is the introduction of "noise" in the estimated treatment effect. However, post hoc analyses of several HF trials suggest that the inclusion of ODI in the trial outcome would have yielded generally consistent treatment effects as an outcome without ODI.[5-18] For example, in trials in HFrEF (e.g., PARADIGM, EMPEROR-Reduced [Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction], DAPA-HF, VICTOR), the treatment effect — including point estimates and confidence intervals — on the primary composite outcome was generally consistent with an expanded composite outcome that included oral intensification of HF therapies or diuretics (Table 1).[11,13,15,17] In VICTOR, a trial that did not meet its prespecified primary outcome, the inclusion of ambulatory events in the primary composite outcome would have yielded a statistically significant treatment effect[17] (Table 1). Although the definition of ODI was heterogeneous across these trials, at least in the VICTOR trial, the estimated treatment effect on the expanded composite worsening HF outcome that included ODI remained consistent in sensitivity analyses that restricted ODI dose increments to 50% or greater.[17]

Post hoc analyses of trials in HFmrEF or HFpEF (e.g., DELIVER, FINEARTS-HF) similarly demonstrated that the treatment effect on the primary composite outcome would have remained consistent when employing an expanded composite outcome that included ODI.[7-9,16] In DELIVER, the treatment effect on the expanded composite outcome remained consistent when the analysis was restricted to ODI sustained for 30 days or more.[9] In SUMMIT (Study of Tirzepatide [LY3298176] in Participants with Heart Failure with Preserved Ejection Fraction and Obesity), the treatment effect of tirzepatide on the primary composite outcome that included ODI was consistent, and less precise in the sensitivity analyses that excluded ODI from the outcome.[5]

In addition to considerations for relative risk reduction, the increase in event rates from an expanded composite outcome that includes ODI would also translate to greater absolute risk reduction with treatment (Table 1).

Challenges and Regulatory Considerations

CHALLENGES INCORPORATING ODI IN HF TRIALS

Although there are advantages to incorporating ODI in the primary definition of worsening HF in clinical trials, there are potential challenges, including the lack of a standard definition for ODI (Fig. 1 and Table 1). Some trials restricted the definition of ODI to the use of loop diuretics, whereas others included additional diuretic classes. Although the inclusion of broader classes of diuretics in the definition captures more events, some diuretics (e.g., thiazide diuretics) have broad indications, and their addition may not be specific for HF. This challenge could be resolved by requiring symptoms of worsening HF in the definition of ODI and requiring the addition of such agents to loop diuretics, aligned with how they were tested in HF trials.[19] In addition to diuretic type, there has been heterogeneity in the magnitude and duration of dose intensification required in HF trials evaluating ODI, and consideration should be given to the optimal definition of ODI in this context.

Another potential challenge regarding the definition of ODI is the clinical context of its deployment. For example, diuretic intensification in a patient who has received suboptimal therapy following hospitalization for HF may not represent a new episode of worsening HF. In at least one study evaluating this contextual challenge, defining discrete ODI events using a 7-day or 30-day separation from antecedent HF hospitalization did not substantially change the treatment effect estimates.[11] In addition to these contextual considerations, there may be heterogeneity in practice patterns across trial sites that could influence the threshold for ODI deployment. Such contextual considerations also apply to HF hospitalization or ambulatory visits with IV diuretic use as end points. Future consideration is warranted to account for this potential heterogeneity.

Investigators may also be concerned about the ability to reliably capture episodes of ODI. For example, if patients do not report diuretic self-adjustments at a clinic visit, events may be underestimated. Structured trial documentation templates, medication reconciliation processes, and patient-recorded medication changes could close this gap. The feasible capture of outpatient ODI in contemporary trials — with documented increases in event rates and consistency of treatment effect — is generally reassuring.[6-18]

REGULATORY CONSIDERATIONS FOR INCLUDING ODI IN HF TRIALS

Regulators may express hesitation regarding the approval of therapies evaluated using end points of limited relevance. Unlike surrogate end points that are markers of disease activity, we argue that ODI is a clinically and prognostically relevant worsening HF event, and that its targeting may influence other clinical events, such as HF hospitalization and death.

PARADISE-MI (Prospective ARNI versus ACE inhibitor Trial to Determine Superiority in Reducing Heart Failure Events after Myocardial Infarction)[18] established a precedent in this regard by receiving regulatory approval for the inclusion of ODI in its primary outcome. Following discussions between trial investigators and regulators, the trial received approval for the inclusion of ambulatory HF in the primary composite outcome, the inclusion of ODI in the definition of ambulatory HF, and the definition of ODI itself (Fig. 1). The discussion was guided by data from a post hoc analysis of the PARADIGM-HF trial that demonstrated increased risk of cardiovascular death following outpatient HF events. Looking ahead, primary composite outcomes that include ODI could be analyzed using hierarchical analytical approaches that prioritize HF hospitalizations over ambulatory HF events, as is specified in the ongoing ELEVATE-HFpEF trial (Randomized Trial of ELEVATEd Cardiac Pacing Rate for Personalized Treatment of Heart Failure with Preserved Ejection Fraction; NCT06678841). In the case of ELEVATE-HFpEF, regulators approved a definition of ODI similar to the standardized definition we propose here (Fig. 2) and further approved a hierarchical analytical plan that ranked ODI below HF hospitalization in the primary analysis. These discussions were informed by a review of data that demonstrated the prognostic relevance of ODI, as highlighted above. Future trials may consider sensitivity analyses to assess event rates and treatment effect estimates using different definitions of ODI.

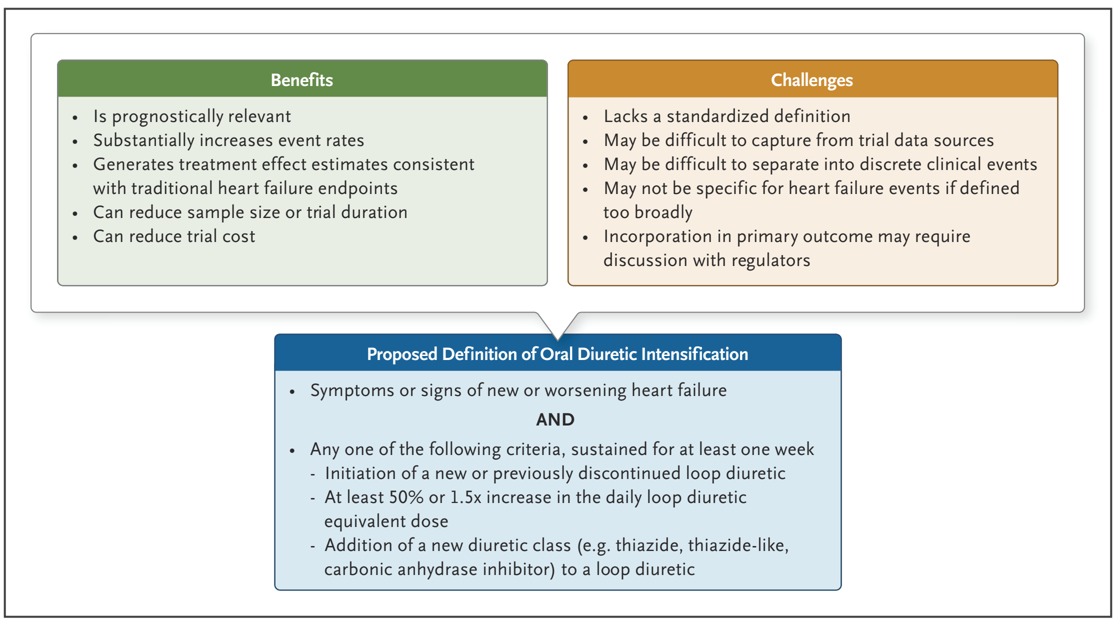

Figure 2. Oral Diuretic Intensification as a Worsening Heart Failure Event in the Primary Outcome of Clinical Trials.

HF denotes heart failure.

A Standardized Definition

As a starting point for further elaboration and with the intent of optimizing specificity for a true HF event, we propose a definition of ODI that requires symptoms or signs of new or worsening HF with at least one of three additional criteria sustained for at least 1 week (Fig. 2): (1) the initiation or reinitiation of a loop diuretic; (2) at least 50% or 1.5-fold increase in daily loop-diuretic–equivalent dose; or (3) the addition of a new diuretic class to a loop diuretic. We acknowledge that there is no harmonized definition for ODI and that thresholds of definition may be guided by practical considerations, such as reliable recall, extraction from pharmacy records, and capture in data forms. Post hoc analyses in HF trials have generally suggested that the degree of diuretic intensification was associated with the magnitude of mortality risk.[11] Further work is needed to evaluate the potential trade-offs of sensitivity and specificity for numeric thresholds and duration of intensification in the definition of ODI.

Conclusion

In clinical practice, worsening HF is commonly managed with ODI. To date, the definition of ODI has varied across HF pharmacotherapy trials. In post hoc analyses of such trials, ODI is associated with a higher risk of mortality, and limited available data suggest that there is a graded association between the magnitude of ODI intensification and mortality risk. In this article, we propose that an expanded definition of worsening HF that includes ODI in the primary outcome could increase event rates and yield consistent treatment effect estimates in trials, improving statistical efficiency. We therefore suggest that the definition of worsening HF in trials be expanded from its current formulation (i.e., HF hospitalization and urgent HF visits with ambulatory IV therapies) to a definition that includes ODI.

Disclosures

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References

1. Van Spall HGC, Averbuch T, Damman K, Voors AA. Risk and risk reduction in trials of heart failure with reduced ejection fraction: absolute or relative? Eur J Heart Fail 2021;23:1437-1444. DOI: 10.1002/ejhf.2248.

2. Hicks KA, Mahaffey KW, Mehran R, et al. 2017 cardiovascular and stroke endpoint definitions for clinical trials. Circulation 2018;137:961-972. DOI: 10.1161/CIRCULATIONAHA.117.033502.

3. Ambrosy AP, Parikh RV, Sung SH, et al. Analysis of worsening heart failure events in an integrated health care system. J Am Coll Cardiol 2022;80:111-122. DOI: 10.1016/j.jacc.2022.04.045.

4. Zile MR, Bennett TD, St John Sutton M, et al. Transition from chronic compensated to acute decompensated heart failure: pathophysiological insights obtained from continuous monitoring of intracardiac pressures. Circulation 2008;118:1433-1441. DOI: 10.1161/CIRCULATIONAHA.108.783910.

5. Packer M, Zile MR, Kramer CM, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. N Engl J Med 2025;392:427-437. DOI: 10.1056/NEJMoa2410027.

6. Docherty KF, Jhund PS, Anand I, et al. Effect of dapagliflozin on outpatient worsening of patients with heart failure and reduced ejection fraction: a prespecified analysis of DAPA-HF. Circulation 2020;142:1623-1632. DOI: 10.1161/CIRCULATIONAHA.120.047480.

7. Ferreira JP, Liu J, Claggett BL, et al. Outpatient diuretic intensification as endpoint in heart failure with preserved ejection fraction trials: an analysis from TOPCAT. Eur J Heart Fail 2022;24:378-384. DOI: 10.1002/ejhf.2376.

8. Chatur S, Vaduganathan M, Claggett BL, et al. Outpatient worsening among patients with mildly reduced and preserved ejection fraction heart failure in the DELIVER trial. Circulation 2023;148:1735-1745. DOI: 10.1161/CIRCULATIONAHA.123.066506.

9. Cunningham JW, Chatur S, Claggett BL, et al. Finerenone and outpatient worsening heart failure with mildly reduced or preserved ejection fraction: a secondary analysis of the FINEARTS-HF randomized clinical trial. JAMA Cardiol 2025;10:370. DOI: 10.1001/jamacardio.2025.0016.

10. Fontana M, Maurer MS, Gillmore JD, et al. Outpatient worsening heart failure in patients with transthyretin amyloidosis with cardiomyopathy in the HELIOS-B trial. J Am Coll Cardiol 2025;85:753-761. DOI: 10.1016/j.jacc.2024.11.015.

11. Okumura N, Jhund PS, Gong J, et al. Importance of clinical worsening of heart failure treated in the outpatient setting: evidence from the prospective comparison of ARNI with ACEI to determine impact on global mortality and morbidity in heart failure trial (PARADIGM-HF). Circulation 2016;133:2254-2262. DOI: 10.1161/CIRCULATIONAHA.115.020729.

12. Chatur S, Vaduganathan M, Fletcher RA, et al. Canagliflozin reduces oral loop diuretic intensification in patients with type 2 diabetes: a participant-level pooled analysis of the CANVAS and CREDENCE trials. Eur J Heart Fail 2025;27:994-1002. DOI: 10.1002/ejhf.3586.

13. Packer M, Anker SD, Butler J, et al. Effect of empagliflozin on the clinical stability of patients with heart failure and a reduced ejection fraction: the EMPEROR-reduced trial. Circulation 2021;143:326-336. DOI: 10.1161/CIRCULATIONAHA.120.051783.

14. Packer M, Anker SD, Butler J, et al. Cardiovascular and renal outcomes with empagliflozin in heart failure. N Engl J Med 2020;383:1413-1424. DOI: 10.1056/NEJMoa2022190.

15. McMurray JJV, Solomon SD, Inzucchi SE, et al. Dapagliflozin in patients with heart failure and reduced ejection fraction. N Engl J Med 2019;381:1995-2008. DOI: 10.1056/NEJMoa1911303.

16. Pfeffer MA, Claggett B, Assmann SF, et al. Regional variation in patients and outcomes in the treatment of preserved cardiac function heart failure with an aldosterone antagonist (TOPCAT) trial. Circulation 2015;131:34-42. DOI: 10.1161/CIRCULATIONAHA.114.013255.

17. Zannad F, Reddy YNV, Barash I, et al. Effect of vericiguat on worsening heart failure in compensated outpatients with HFrEF: insights from VICTOR. J Am Coll Cardiol 2025 August 27 (Epub ahead of print).

18. Pfeffer MA, Claggett B, Lewis EF, et al. Angiotensin receptor-neprilysin inhibition in acute myocardial infarction. N Engl J Med 2021;385:1845-1855. DOI: 10.1056/NEJMoa2104508.

19. Oskouie, S, Pandey, A, Sauer, A. et al. From hospital to home: evidence-based care for worsening heart failure. JACC Adv 2024;3:101131. DOI: 10.1016/j.jacadv.2024.101131.